

Machine Learning Assignment

1. Implement **Simple Linear Regression manually** without using any ML library.

Dataset

$X = [1, 2, 3, 4, 5]$

$Y = [3, 4, 2, 4, 5]$

Tasks

Calculate:

1. Mean of X ($\bar{X}$)
2. Mean of Y ($\bar{Y}$)
3. Slope (m)
4. Intercept (c)

Expected Output Example

Mean of X = 3

Mean of Y = 3.6

Slope (m) = 0.4

Intercept (c) = 2.4

Regression Equation:

$Y = 0.4X + 2.4$

Predicted Y for X = 6 : 4.8

2. Using the **same dataset from above question**, calculate model performance.

Tasks

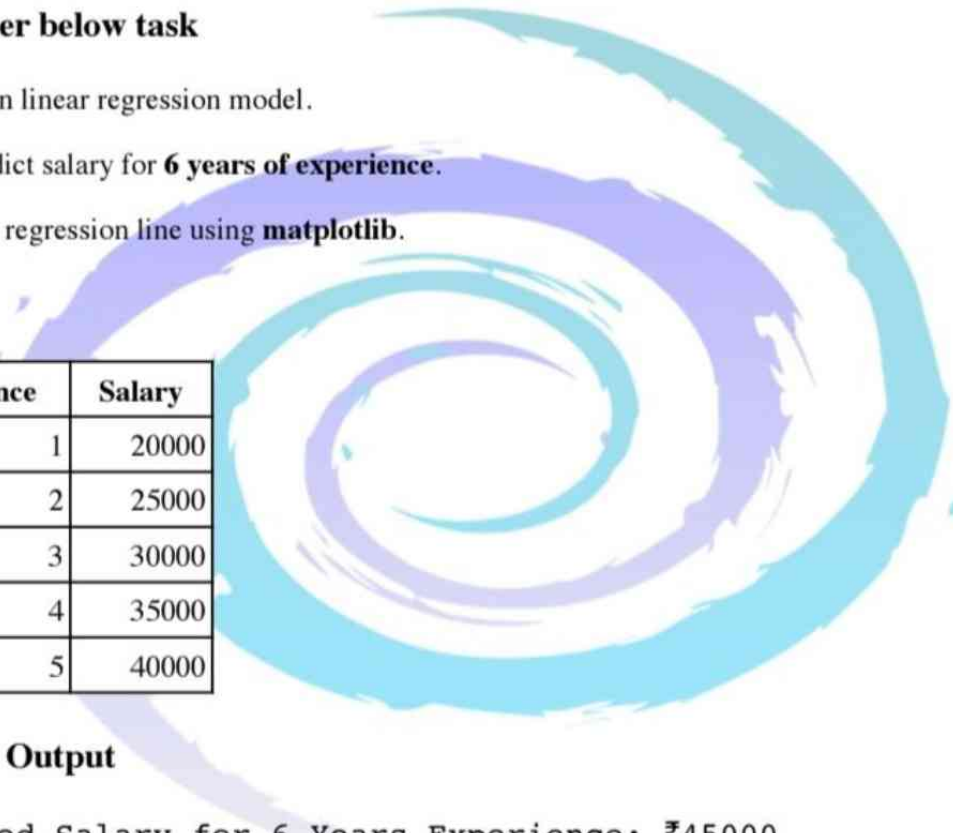
1. Predict all Y values using regression equation.
2. Calculate:
 - Mean Squared Error (MSE)
 - R^2 Score

Show **all intermediate calculations**.

3. Consider below task

1. Train linear regression model.
2. Predict salary for **6 years of experience**.
3. Plot regression line using **matplotlib**.

Dataset



Experience	Salary
1	20000
2	25000
3	30000
4	35000
5	40000

Expected Output

Predicted Salary for 6 Years Experience: ₹45000

Graph should display:

- Data points
- Regression line

4. Why is KNN called a **lazy learner**?
5. What happens if **K is too small**?
6. What happens if **K is too large**?
7. Why does linear regression minimize **squared error**?
8. What is the difference between **MSE and R^2** ?
9. Why **R^2** cannot be greater than 1?
10. Can **KNN** be used for regression?

